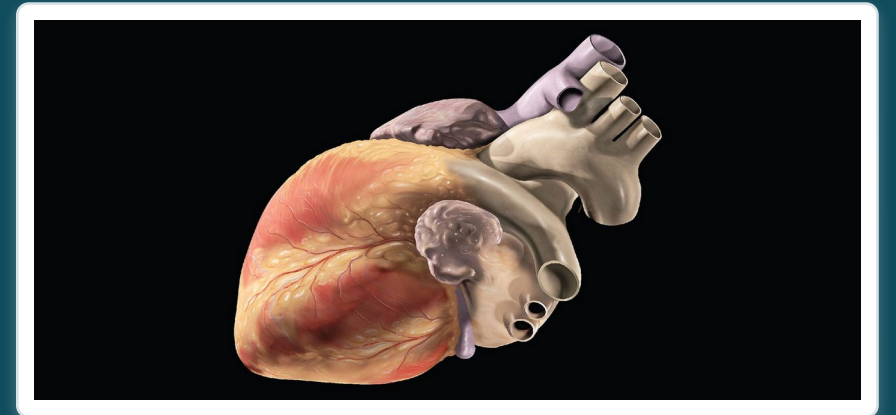


CHAPTER 19

The Cardiovascular System: The Heart

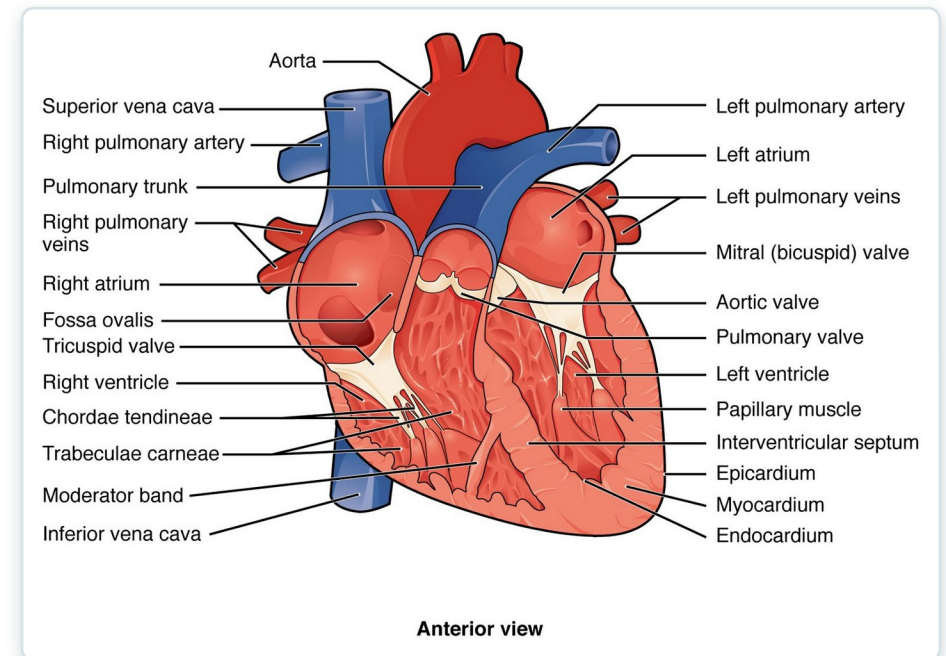
Heart anatomy, the heartbeat, and cardiac function.



Learning Objectives

After studying this chapter, students will be able to:

- 1 Identify and describe the interior and exterior parts of the human heart
- 2 Describe the path of blood through the cardiac circuits
- 3 Describe the size, shape, and location of the heart
- 4 Compare cardiac muscle to skeletal and smooth muscle
- 5 Explain the cardiac conduction system
- 6 Describe the process and purpose of an electrocardiogram



The internal anatomy of the heart.



19.1

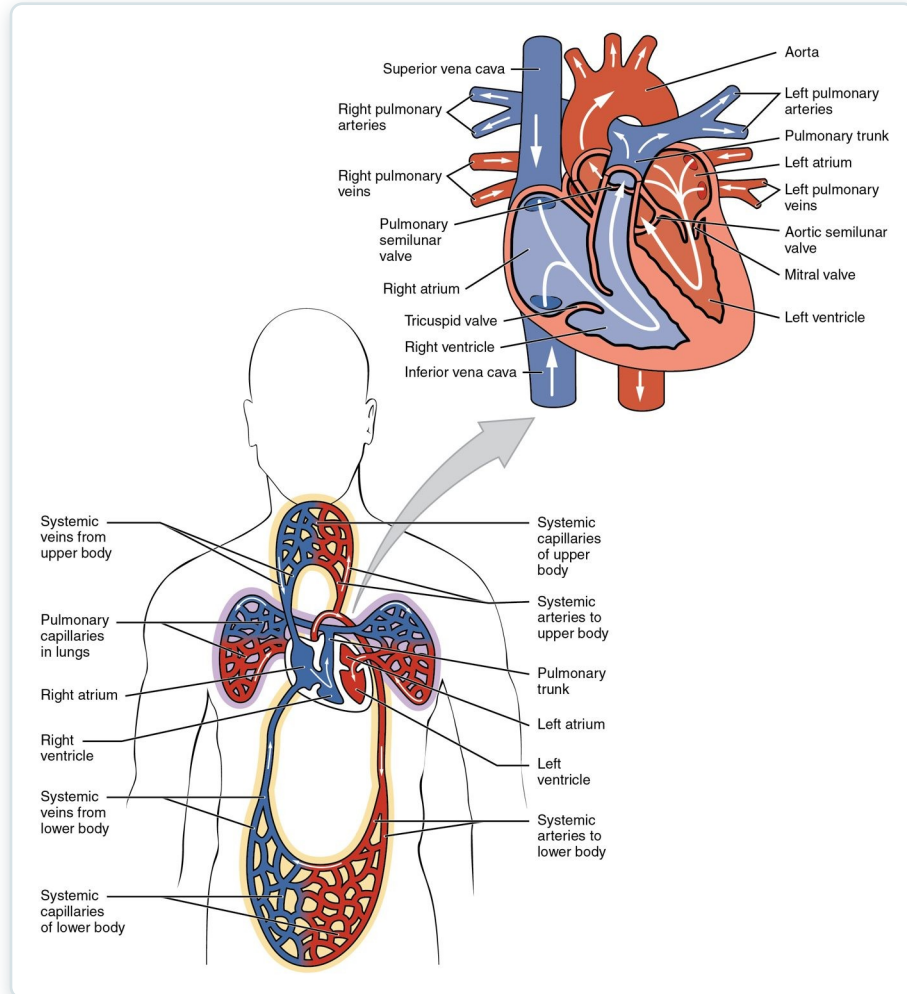
SECTION

Heart Anatomy

The chambers, valves, vessels, and wall of the heart.

SECTION 19.1

Pulmonary & Systemic Circulation



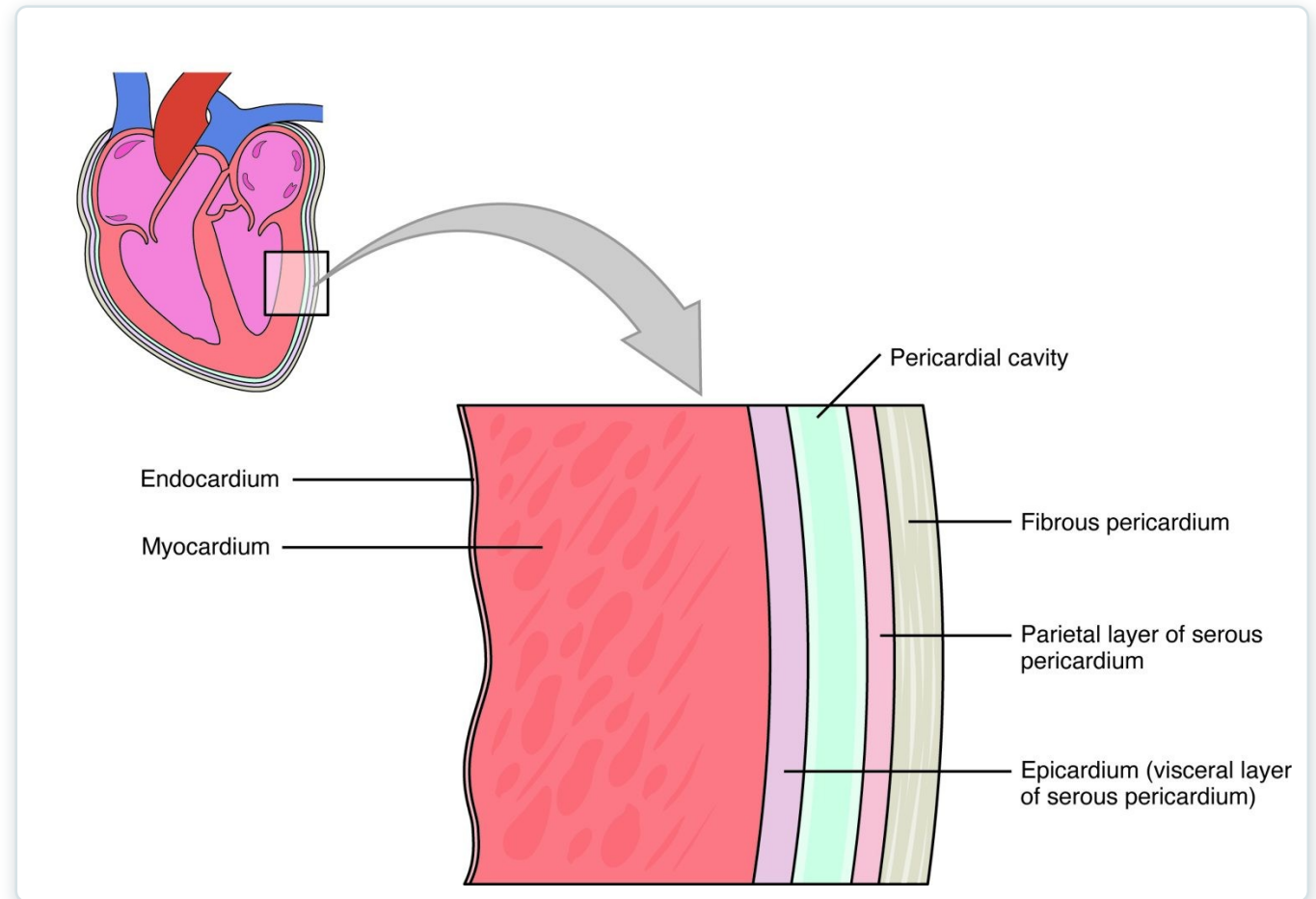
The pulmonary and systemic circuits

- The heart is a double pump
- Right side → lungs (pulmonary)
- Left side → body (systemic)
- Keeps the two circuits separate

SECTION 19.1

The Heart Wall

- Three layers
- Endocardium lines the inside
- Myocardium is the thick muscle
- Pericardium surrounds the heart

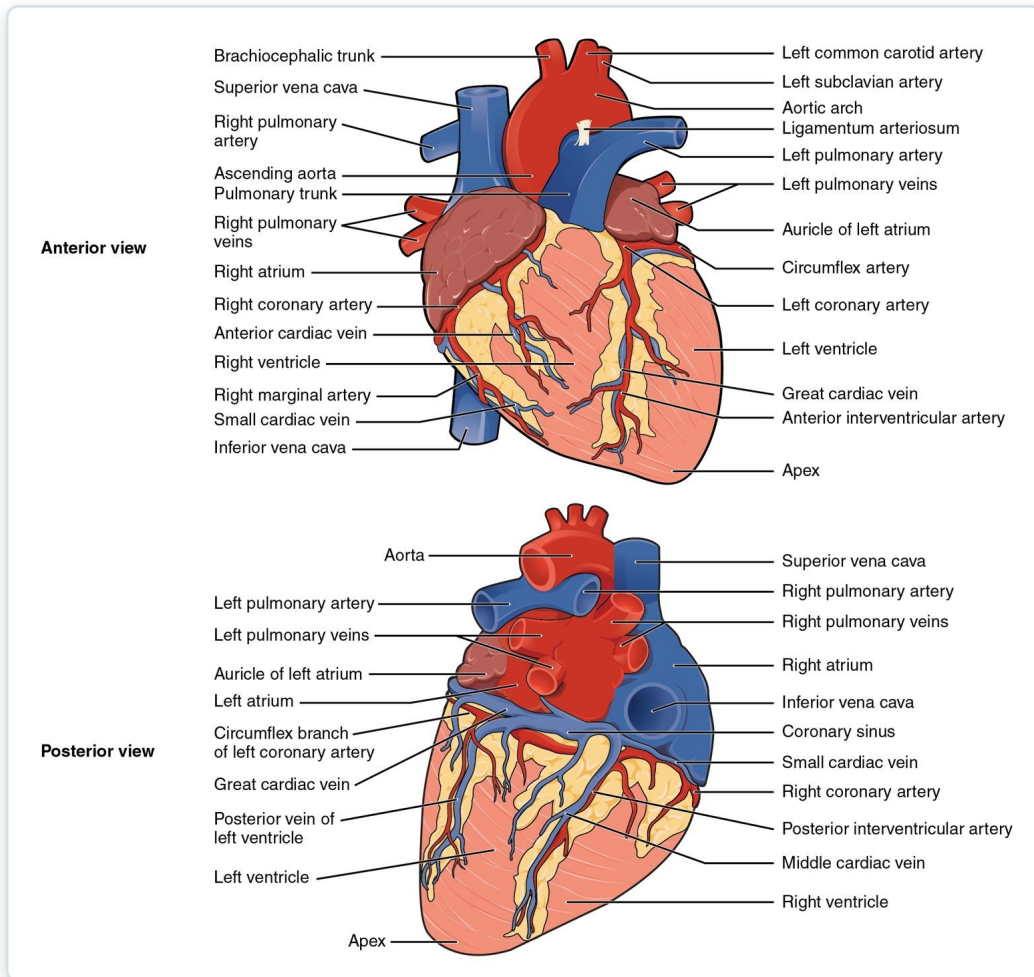


Layers of the heart wall

SECTION 19.1

External Anatomy

- Four chambers: two atria, two ventricles
- Great vessels enter and leave the top
- Coronary vessels on the surface
- Apex points down and to the left

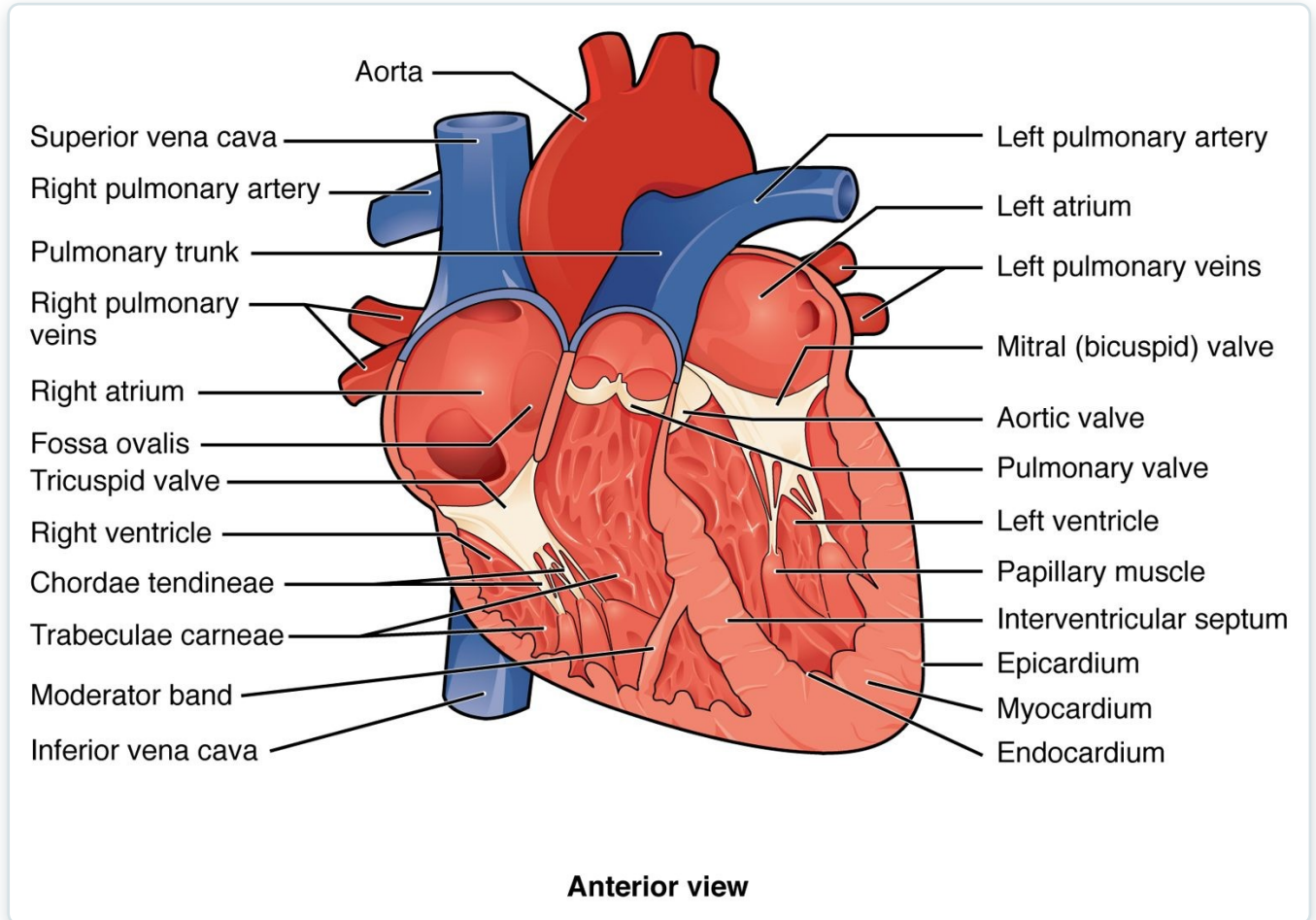


External anatomy of the heart

SECTION 19.1

Internal Anatomy

- Four chambers separated by valves
- Atria receive; ventricles pump
- Valves keep blood moving one way
- Septum divides left and right

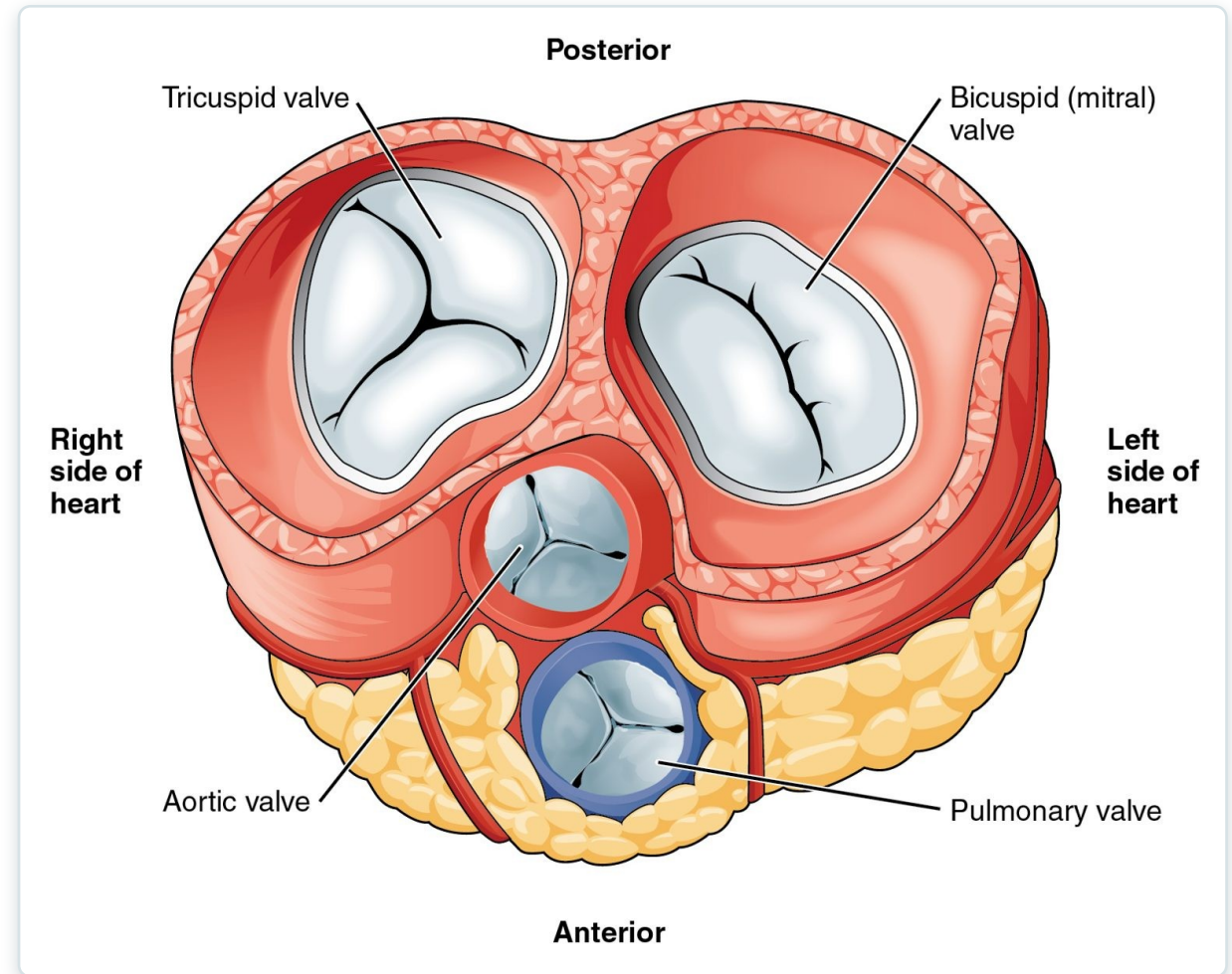


Internal anatomy of the heart

SECTION 19.1

The Heart Valves

- Four valves keep blood moving forward
- Two AV valves (tricuspid, mitral)
- Two semilunar valves (aortic, pulmonary)
- Open and close with pressure changes

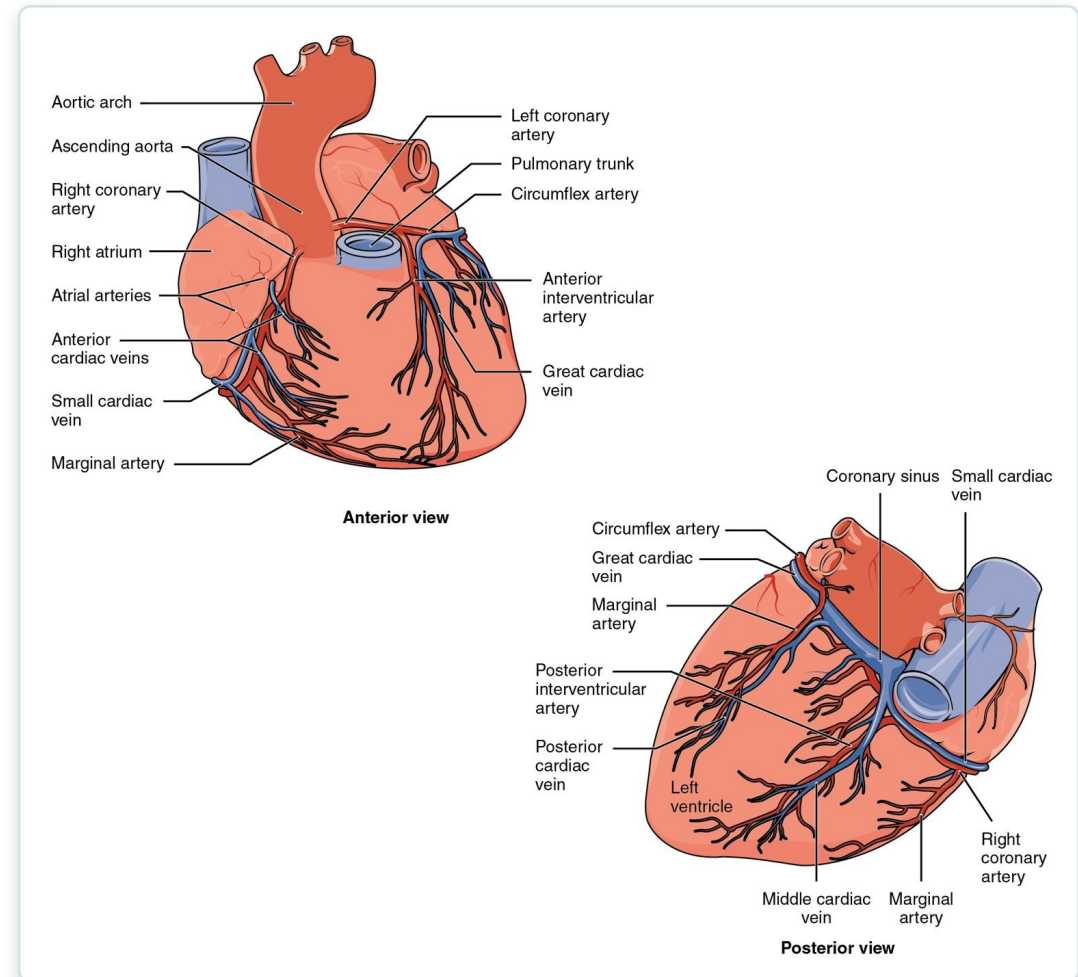


The four heart valves

SECTION 19.1

Coronary Circulation

- The heart needs its own blood supply
- Coronary arteries branch from the aorta
- They feed the myocardium
- Blockage causes a heart attack



The coronary circulation



19.2

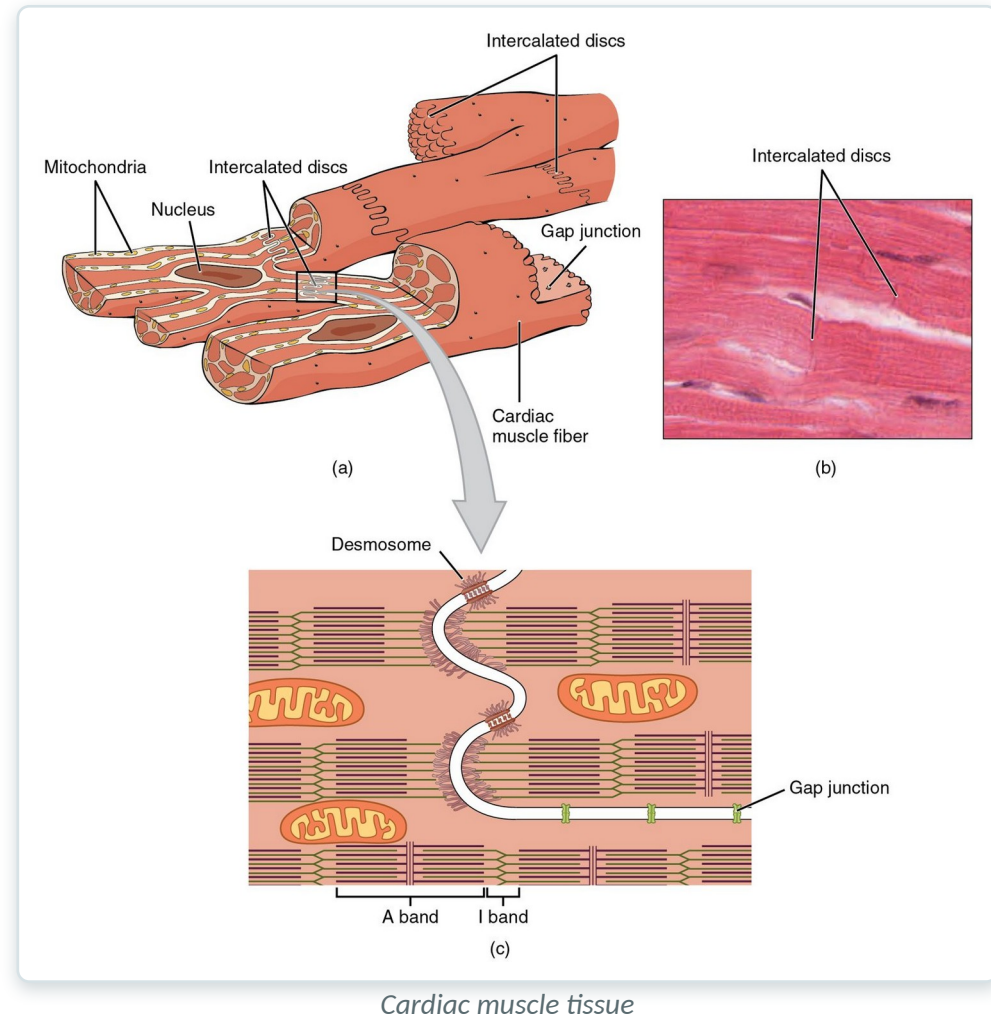
SECTION

Cardiac Muscle and Electrical Activity

Cardiac muscle and the heart's electrical system.

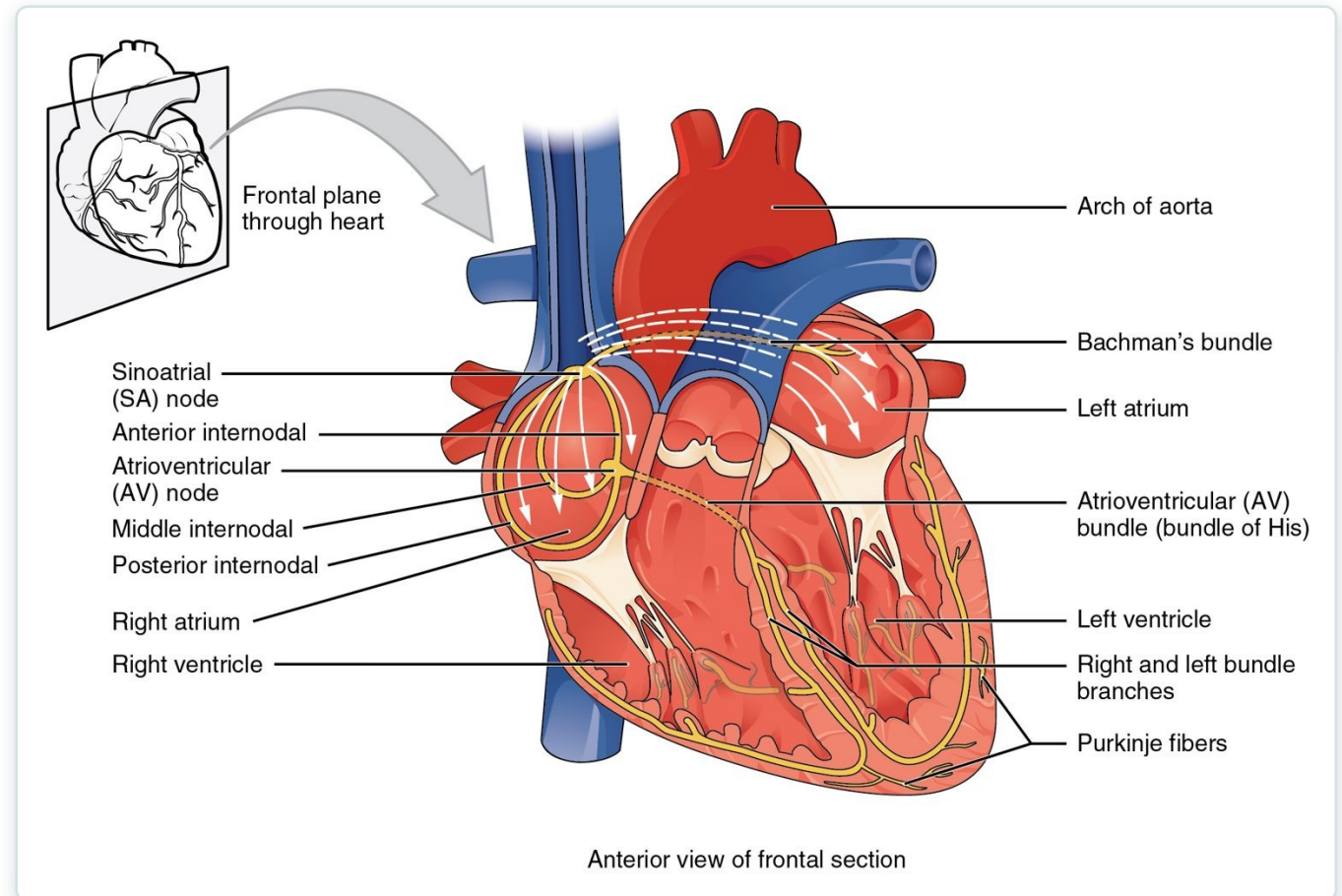
Cardiac Muscle

- Striated but involuntary
- Cells branch and interconnect
- Joined by intercalated discs
- Gap junctions synchronize the beat



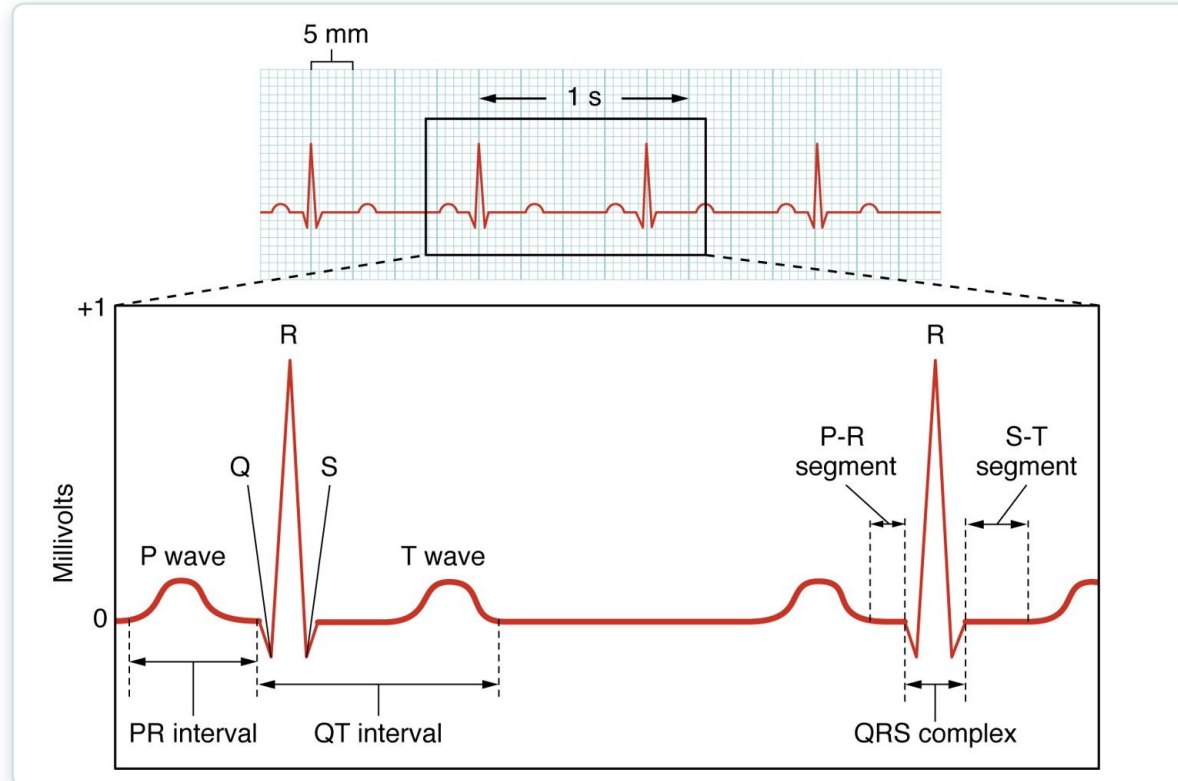
The Conduction System

- The heart sets its own rhythm
- SA node is the pacemaker
- AV node delays the signal
- Bundle and Purkinje fibers spread it



The cardiac conduction system

The Electrocardiogram (ECG)

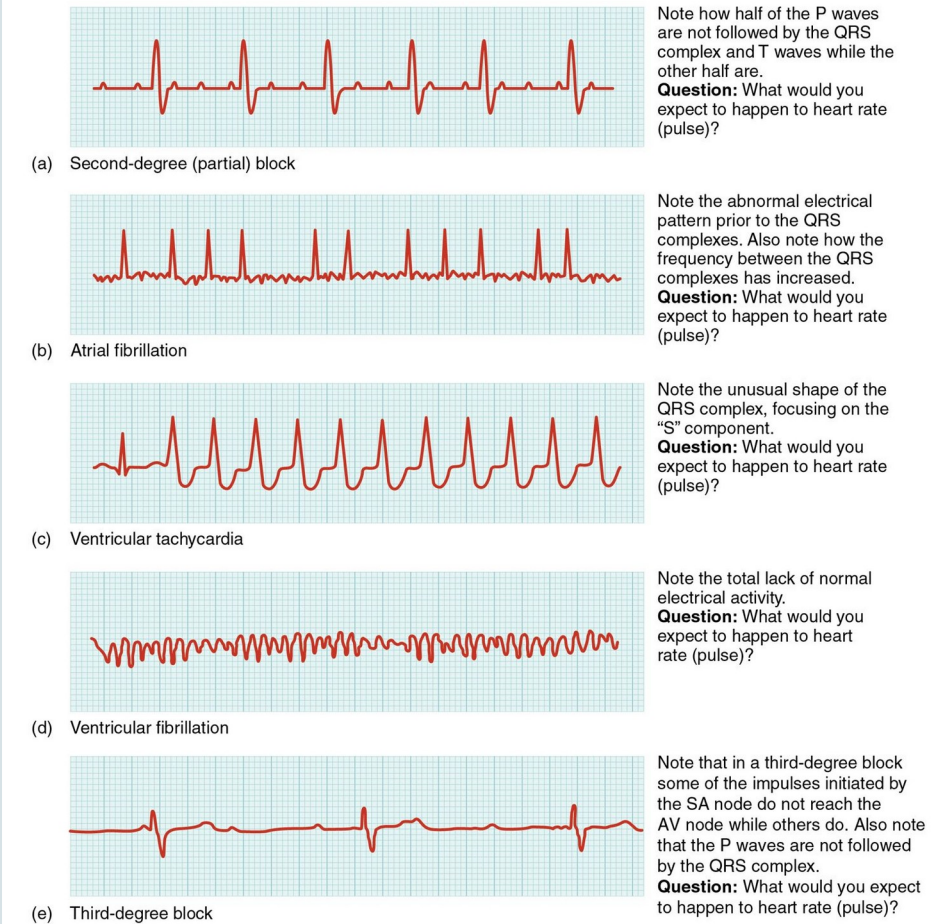


A normal ECG tracing

- Records the heart's electrical activity
- P wave = atrial depolarization
- QRS = ventricular depolarization
- T wave = ventricular repolarization

Arrhythmias

- Abnormal heart rhythms
- Show as abnormal ECG patterns
- Atrial fibrillation, V-tach, V-fib
- Some are life-threatening



Examples of cardiac arrhythmias

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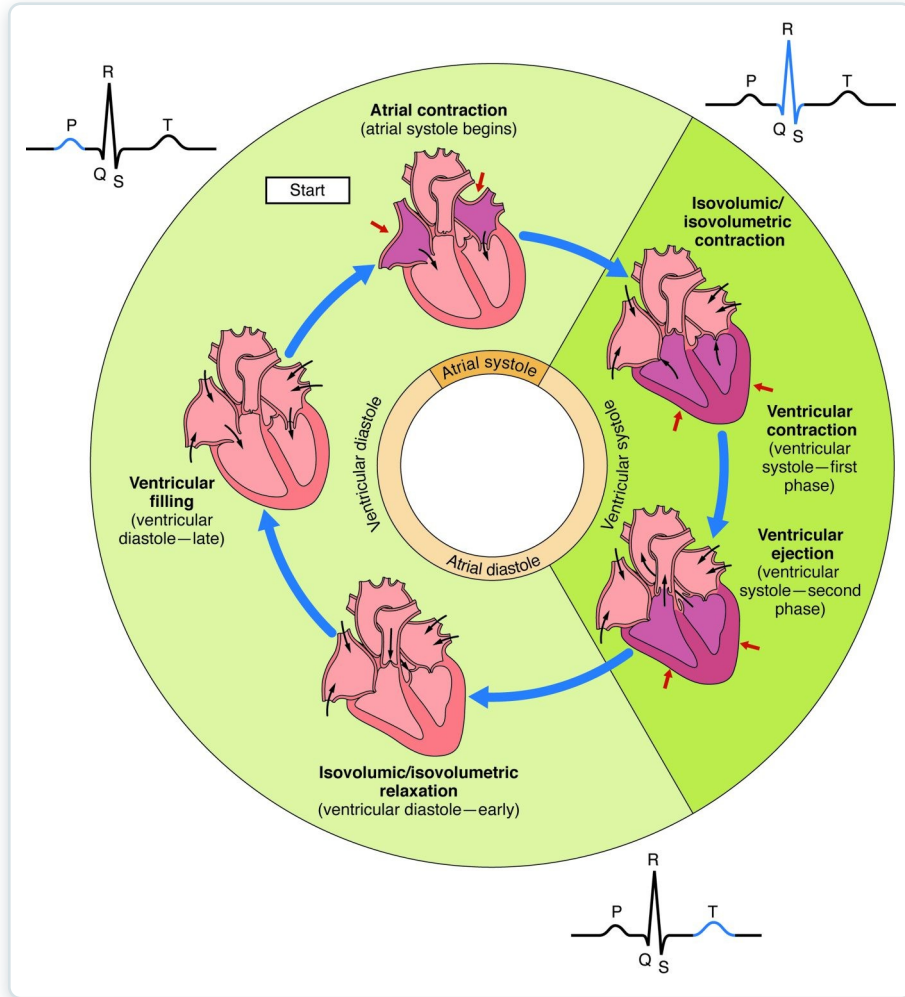
19.3

SECTION

Cardiac Cycle

The cardiac cycle — one complete heartbeat.

The Cardiac Cycle

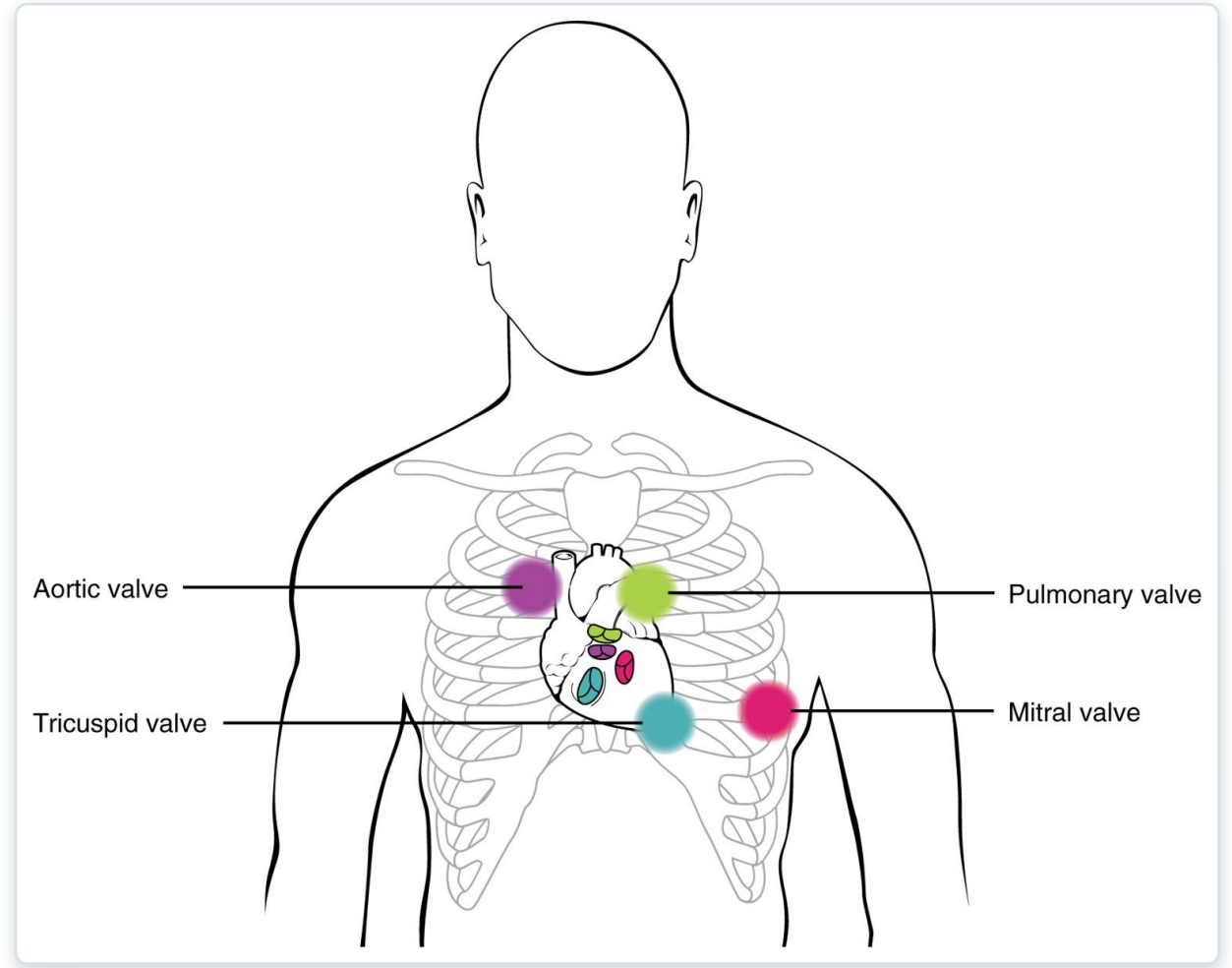


The cardiac cycle

- One heartbeat = one cardiac cycle
- Atria contract, then ventricles
- Systole = contraction
- Diastole = relaxation

Heart Sounds

- Valves closing make the sounds
- "Lub" = AV valves closing
- "Dup" = semilunar valves closing
- Heard at specific chest sites



Auscultation points for heart sounds



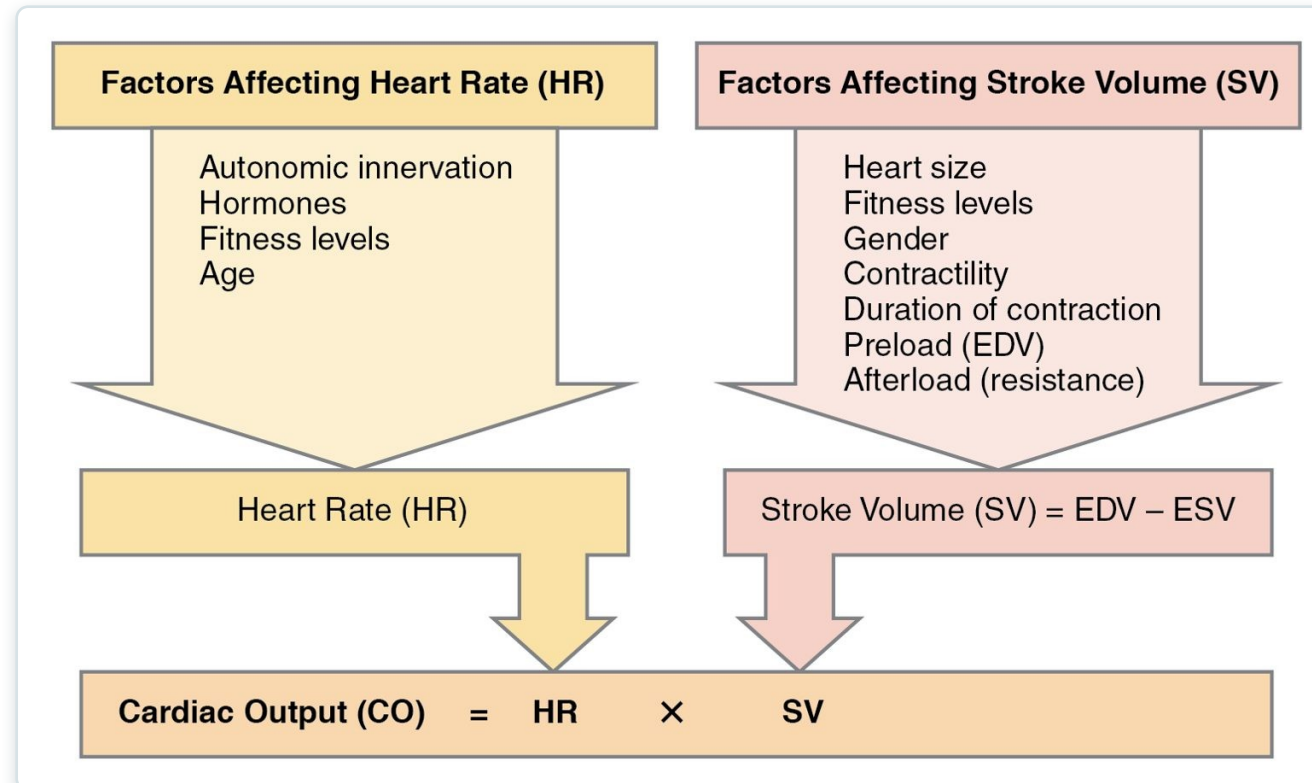
19.4

SECTION

Cardiac Physiology

Cardiac output and how it is regulated.

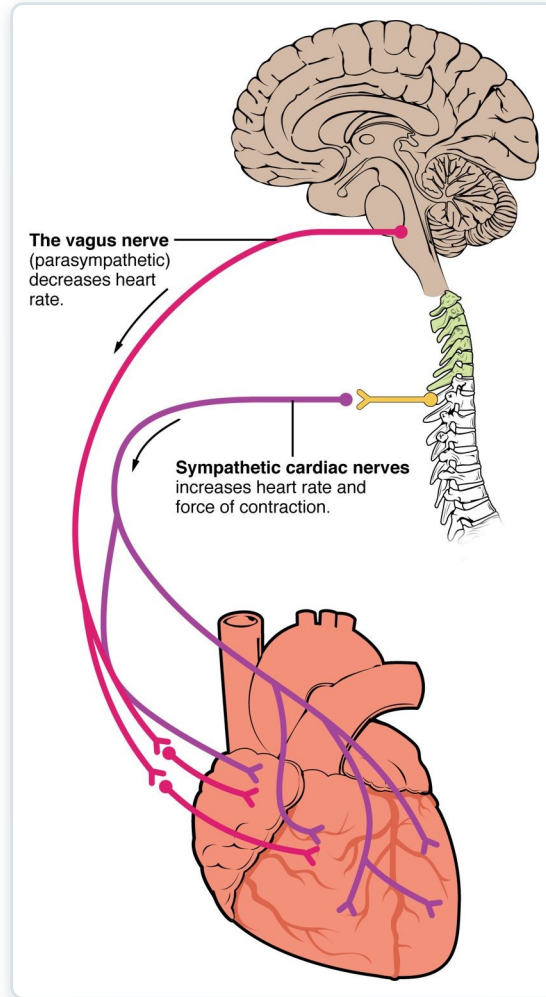
Cardiac Output



Cardiac output: heart rate and stroke volume

- Output = heart rate × stroke volume
- Stroke volume = blood per beat
- About 5 liters per minute at rest
- Rises with exercise and demand

Autonomic Control of the Heart



Autonomic control of the heart

- The brain adjusts heart rate
- Sympathetic input speeds it up
- The vagus nerve slows it down
- Matches output to demand



19.5

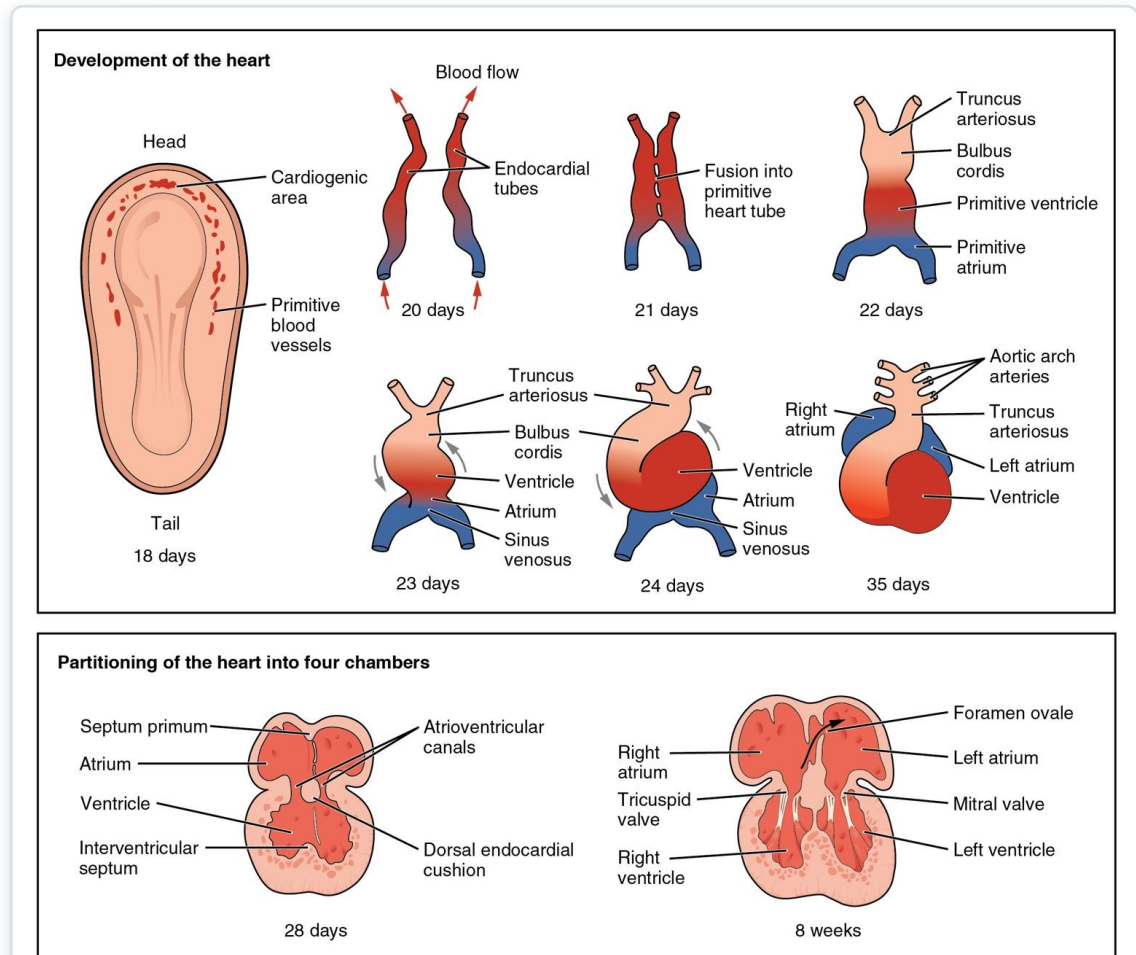
SECTION

Development of the Heart

How the heart develops before birth.

Development of the Heart

- The first organ to function
- Begins as a simple tube
- Folds and divides into four chambers
- Defects can occur in the process



Development of the heart

Key Takeaways

- The heart is a double pump driving the pulmonary and systemic circuits.
- Four chambers and four valves keep blood flowing in one direction.
- Cardiac muscle and the conduction system generate a coordinated, automatic heartbeat.
- The ECG records the heart's electrical activity and reveals arrhythmias.
- Cardiac output (rate $\times$ stroke volume) is tuned by the autonomic nervous system.

CLINICAL CONNECTIONS

Why the Heart Matters in Practical Nursing

Heart Attack

A blocked coronary artery starves the myocardium of oxygen.

Heart Failure

A weakened pump can't meet the body's demands.

Arrhythmias

Abnormal rhythms range from harmless to lethal.

Heart Murmurs

Faulty valves produce abnormal heart sounds.

ECG Monitoring

A core tool for assessing cardiac patients.

Vital Signs

Pulse reflects heart rate, rhythm, and strength.